Vincenzo Marco De Luca

PHD STUDENT IN INFORMATION ENGINEERING AND COMPUTER SCIENCE

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RESEARCH SUMMARY

PhD candidate specializing in **Graph Neural Networks, Explainable AI, and Computational Social Science**. Focused on Human-Centered AI applications, human-machine teaming, and surgical team dynamics.

EDUCATION

PhD in Information Engineering and Computer Science

November 2023 – In progress

University of Trento - Department of Information Engineering and Computer Science

- **Scholarship:** AI coaching for human-machine teaming
- **Topics:** Graph Neural Networks, Computational Social Science, AI for Healthcare
- **Advisor:** Prof. Andrea Passerini (University of Trento)
- **Co-advisor:** Prof. Marco Zenati (Harvard University)
- **Co-advisor:** Prof. Giovanna Varni (University of Trento)

Visiting PhD Student

July 2025 – November 2025

Max Planck Institute for Intelligent Systems - Haptic Intelligence Department

- **Supervisor:** Prof. Katherine J. Kuchenbecker

Summer School - OxML (Representation Learning)

July 2024

University of Oxford

Summer School - London on Geometry and Machine Learning (LOGML)

July 2024

Imperial College London

Master's Degree in Artificial Intelligence Systems (Computer Engineering)

September 2020 – July 2023

University of Trento - Department of Information Engineering and Computer Science

- **Mark:** 110/110 cum laude
- **Thesis:** Regularize learning in Graph Neural Networks through Explainability
- **Skills acquired:** Machine Learning and Deep Learning for Language Processing and Computer Vision, Optimization, Logic, Planning, Knowledge Engineering, Quantum Computing

Bachelor's Degree in Computer Science

September 2016 – September 2020

University of Naples Parthenope - Department of Science and Technology

- **Mark:** 110/110 cum laude
- **Thesis:** CLT-DTA: Chemical Language Transformer for Drug Target Binding Affinity

- **Skills acquired:** Full-Stack Developer, Software Engineering, Computer Science theory, Machine Learning, Probability, Statistics, Linear Algebra, Real Algebra, Optimization, Scientific Calculus

ACCEPTED ARTICLES

2026 **From Multimodal Observation to Interpretable Suggestions: Counterfactual Time-Expanded Relational Modeling of Surgical Teams**
Vincenzo Marco De Luca, Antonio Longa, Giovanna Varni, Andrea Passerini
ACM Multimedia (ACM MM 2026)

2026 **Towards Actionable Surgical Team Dynamics: from Teamwork to Counterfactual Annotations**
Vincenzo Marco De Luca, Antonio Longa, Andrea Passerini
HCOMP '26: Proceedings of the 2026 ACM Conference on Human-AI Complementarity and Alignment

2026 **Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs**
Vincenzo Marco De Luca, Antonio Longa, Giovanna Varni, Andrea Passerini
Proceedings of the 5th International Conference on Hybrid Human-Artificial Intelligence, Frontiers in Artificial Intelligence and Applications, Sage Journals

2026 **Overcoming Shortcut Learning in Graph Neural Networks through Active Explanation Guidance**
Taraneh Younesian, Steve Azzolin, Antonio Longa, Francesco Ferrini, Vincenzo Marco De Luca, Stefano Teso
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2026) - Workshop on Mining and Learning with Graphs (MLG 2026)

2026 **Boosting Team-Modeling through Tempo-Relational Representation**
Vincenzo Marco De Luca, Giovanna Varni, Andrea Passerini
Springer Nature - Cognitive Computation

2024 **XAI-Drop: Don't Use What You Cannot Explain**
Vincenzo Marco De Luca, Antonio Longa, Pietro Liò, Andrea Passerini
Proceedings of the Third Learning on Graphs Conference, Proceedings of Machine Learning Research (PMLR)

2021 **Chemical Language Transformer for Drug-Target Binding Affinity**
Vincenzo Marco De Luca, Antonino Staiano, Alessio Ferone
17th International Conference on Computational Intelligence Methods for Bioinformatics and Biostatistics (CIBB 2021)

UNDER SUBMISSION

2026 **Graph Additive Models disentangling feature, topology, and interactions**
Clara Alvarez-Rodriguez, Steve Azzolin, Vincenzo Marco De Luca, Filippo Maria Bianchi, Andrea Passerini, Antonio Longa

ORAL, TALK, AND POSTER PRESENTATIONS

2026 Poster **From Multimodal Observation to Interpretable Suggestions: Counterfactual Time-Expanded Relational Modeling of Surgical Teams** – ACM Multimedia (ACM MM 2026)

2026 Oral **Towards Socially-Aware Modeling of Surgical Teams via Time-Expanded Graphs** – Conference on Complex Systems (CCS 2026)

2026 Research Seminar **Towards Actionable Surgical Team Dynamics: from Teamwork Inference to Actionable Feedback** – Harvard Medical School

2026 Oral **Towards Actionable Surgical Team Dynamics: from Teamwork to Counterfactual Annotations** – ACM Conference on Human-AI Complementarity (ACM HCOMP 2026)

2026 Oral Poster **Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs** – International Conference Series on Hybrid Human-Artificial Intelligence (HHAI 2026)

2026 Oral Poster **Actionable Real-Time Modeling of Surgical Team Dynamics via Time-Expanded Interaction Graphs** – ECML PKDD 2026 - Workshop on Hybrid Human-Machine Learning and Decision Making (HLDM 2026)

2026 Poster **Time-Expanded Graph Learning for Multi-Agent Temporal Interaction Modeling with Counterfactual Interventions** – Learning on Graphs - Pisa Meetup 2026

2025 Research Seminar **Tempo-Relational Neural Networks for Advancing Team Modeling in the Operating Room** – Max Planck Institute for Intelligent Systems

2025 Poster **Boost Human-Machine Teams performance through Tempo-Relational World Representation** – ICT Days - Trento

2024 Oral Poster Best Poster Award **XAI-Drop: Don't use what you cannot explain** – Learning on Graphs - Siena Meetup 2024

2024 Poster **XAI-Drop: Don't Use What You Cannot Explain** – Learning on Graphs 2024

2024 Oral Poster **XAI-Drop: Don't Use What You Cannot Explain** – Learning on Graphs - Amsterdam Meetup 2024

2024 Oral Poster **XAI-Drop: Don't Use What You Cannot Explain** – Learning on Graphs - Paris Meetup 2024

2023 Poster **XAI-based Regularizer for Graph Neural Networks** – Learning on Graphs - Trento Meetup 2023

2024 Oral Poster **XAI-Drop: Don't Use What You Cannot Explain** – Future Artificial Intelligence Research (FAIR) General Conference 2024

2021 Oral **Chemical Language Transformer for Drug-Target Binding Affinity** – 17th International Conference on Computational Intelligence Methods for Bioinformatics and Biostatistics (CIBB 2021)

TEACHING AND TUTORING

Teaching Assistant **Introduction to Web Programming** 2024 – In progress

Bachelor in Computer Science - University of Trento

Tutor **Scientific Programming** 2024 – 2025

Master in Data Science - University of Trento

Tutor **Scientific Programming** 2024 – 2025

Master in Quantitative and Computational Biology - University of Trento

Tutor **Computer Programming** 2024 – 2025

Bachelor in Information Engineering - University of Trento

Teacher **Neural Networks** 2024 – 2025

Scientific High School Galileo Galilei - Trento

Tutor **Computer Programming** 2022 – 2023

Bachelor in Information Engineering - University of Trento

Teacher **Computer Programming** 2022 – 2023

High School - I.T.E.T. Pilati - Cles, Trento

Teacher **Computer Science for Teachers** 2022 – 2023

Teacher Dojo and FabLab - Trento

Private Teacher **in Computer Science** 2021 – 2024

Superprof

ADDITIONAL ACADEMIC EXPERIENCES

- **2024 – In progress:** Program Committee - ACM Conference on Intelligent User Interfaces (IUI).
- **2024 – In progress:** Reviewer: LOG 2025, alt.CHI 2025, IUI 2025, IJCNN 2026, ACM MM 2026, IUI 2026.
- **2025:** Organizing Committee member - London on Geometry and Machine Learning (LOGML) 2025.
- **2023:** Internship - Geometric Deep Learning research - University of Trento (Structured Machine Learning Lab - Prof. Andrea Passerini).
- **2020:** Internship - Deep Learning for Bioinformatics - University of Naples Parthenope (High Performance Scientific Computing Smart-Lab - Prof. Antonino Staiano).
- **2019:** Venue Technology Manager - Universiadi Napoli 2019.
- **2018 – 2019:** Software Developer - iOS Foundation Napoli.

LANGUAGES

Language	Listening	Reading	Spoken Interaction	Spoken Production	Writing
Italian	Mother tongue				
English	C1	C1	C1	C1	C1
Spanish	B2	B2	B2	B2	B2
French	B1	B1	B1	B1	B1

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Vincenzo Marco De Luca